

RESEARCH PAPER

TattooAudit: A Benchmark-Grounded Interactive Expert-Review System for Open-Set Tattoo Analysis

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Abstract. In forensic and other public-interest image-analysis contexts, inaccurate or incomplete semantic descriptions of tattoos can misdirect case retrieval, annotation review, and subsequent human interpretation. Existing open-set tattoo benchmarks summarize average model performance, but they provide limited support for identifying cases that require expert attention, explaining why the system prioritized them, or determining what action should follow inspection. We address this gap by combining a controlled benchmark with an interactive expert-review system. We evaluate Gemma3:12B, Qwen2.5-VL-7B, and LLaMA 3.2 Vision 11B on TSSD2023 under oracle-segmentation conditions. For each case, we evaluate the original image and class-specific, ground-truth-derived crops rendered on black and white backgrounds. An offline pipeline generates the crops, executes the VLMs, normalizes their outputs, aggregates crop-level predictions at the image level, and computes case-level metrics. The interactive workspace does not generate masks, create crops, or invoke the VLMs during review; it consumes the resulting precomputed artifacts. We evaluate multi-label semantic naming, open-set behavior, overprediction, omission, and cross-background stability. We then introduce *TattooAudit*, a benchmark-grounded expert-review workspace that transforms precomputed experimental outputs into a transparent review workflow. The system provides a filterable review queue, case-level visual evidence, cross-model comparisons, explicit prioritization indicators, structured expert actions, decision history, and exportable records. The reference annotation remains authoritative: model outputs provide supporting evidence, and the system stores proposed revisions separately rather than overwriting the benchmark. Qwen2.5-VL-7B achieves the strongest overall balance of micro-F1, runtime, and robustness. Gemma3 outputs the *unknown* token more frequently on isolated crops, whereas LLaMA substantially overpredicts under crop conditions. We do not claim operational validation or a completed usability evaluation. Instead, we contribute a functional research artifact and a technically explicit workflow that converts benchmark discrepancies into structured, reviewable, and traceable opportunities for expert decision-making.

Keywords: vision–language models, open-set recognition, tattoo analysis, interactive expert review, human-in-the-loop, decision support, public-interest AI

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1 Introduction

Tattoos can provide soft biometric evidence when conventional identifying cues are unavailable, degraded, or inconclusive [Ngan and Grother, 2015; Di and Patel, 2016; Han and Jain, 2013]. Previous studies have examined tattoo retrieval, matching, and investigative search, including situations in which only partial visual evidence is available [Ngan and Grother, 2015; Han *et al.*, 2019; Fang *et al.*, 2018]. However, tattoos require careful interpretation because they may reveal personal, cultural, or socially sensitive information beyond that typically exposed by standardized biometric traits [Bacchini and Lorusso, 2018].

In high-stakes public-interest settings, different errors can produce markedly different consequences. An omitted label may prevent the retrieval of a relevant image. An unsupported label may direct attention toward an association that the visual evidence does not support. An overconfident interpretation may also conceal the need for specialist review. Research on tattoo-assisted identification of trafficking victims illustrates why semantic outputs must remain inspectable and contestable rather than operate as autonomous conclusions [Fang *et al.*, 2018]. Our study does not infer identity,

intent, criminal affiliation, ethnicity, or cultural membership from a tattoo. Instead, it examines semantic naming within a bounded benchmark vocabulary and uses case-level discrepancies to support expert review of dataset annotations.

From a computer-vision perspective, tattoos pose several challenges. They are stylized, compositionally dense, and semantically heterogeneous. A single image may contain multiple objects, decorative elements, overlapping strokes, and fragments whose interpretation depends on the surrounding composition [Brilhador *et al.*, 2024]. These properties make a strictly closed-set formulation inadequate for tattoo understanding. Some visual content belongs to known classes, whereas other content falls outside the available vocabulary. Open-set recognition must therefore identify known categories while avoiding the overconfident assignment of unfamiliar content to the nearest known class [Geng *et al.*, 2021; Nunes *et al.*, 2023].

Aggregate benchmark scores do not reveal all case-level failures. A model may perform well on average yet remain unstable on individual cases. It may react to irrelevant background changes, omit reference labels, introduce plausible but unsupported labels, or use the *unknown* token inconsistently. These behaviors matter because an analyst must understand

not only *what* the model predicted, but also *why* the case requires attention and *what* action should follow the inspection.

Brilhador *et al.* [2024] introduced Open-Set Tattoo Semantic Segmentation and the TSSD2023 benchmark, which distinguishes known known classes (KKCs) from unknown unknown classes (UUCs). We use this benchmark to study a different stage of the analysis pipeline. Rather than proposing a segmentation model, we treat the reference masks as oracle segmentation and isolate the semantic-naming task. This design allows us to examine VLM behavior without conflating semantic errors with mask-prediction errors.

We investigate two complementary research questions:

RQ1 – Model behavior.

Once a relevant tattoo region is available, how accurately and consistently do general-purpose VLMs assign semantic labels under original-image and reduced-context conditions?

RQ2 – Interactive review.

How can benchmark outputs be organized into a transparent expert-review workflow that supports triage, evidence inspection, routing, and traceable decisions without treating model predictions as final authority?

To address RQ1, we evaluate Gemma3:12B, Qwen2.5-VL-7B, and LLaMA 3.2 Vision 11B under a unified zero-shot protocol. We present each test instance in three visual conditions: the original image, a ground-truth-derived crop on a black background, and the same crop on a white background. We parse the textual responses into label sets and measure semantic agreement, overprediction, omission, *unknown* usage, and black/white stability.

To address RQ2, we present *TattooAudit*, a functional interactive expert-review system built on the benchmark outputs. *TattooAudit* implements a review loop in which an intended specialist user can: (i) filter and open prioritized cases; (ii) inspect the original image, reference mask, and GT-derived crops; (iii) compare the three VLMs across the three input conditions; (iv) examine the indicators that produced the audit-priority score; (v) confirm the current reference, propose a revision, mark the case as ambiguous, request re-annotation, or exclude the case from a high-confidence pool; and (vi) preserve the decision rationale and review history in a separate record.

The study separates two technical boundaries. The upstream offline evaluation starts from the TSSD2023 original images, reference masks, and reference labels. Our preprocessing uses each mask to generate black- and white-background crops. The three VLMs then process the original image and both derived crops, and the pipeline stores normalized outputs and case-level metrics. The downstream *TattooAudit* interface starts only after these artifacts exist; it supports expert review but does not perform segmentation or VLM inference during the interactive session. Accordingly, the term *original image* denotes one evaluated input condition; it does not imply that the current interface receives only an image and automatically derives every subsequent artifact.

TattooAudit separates three concepts to prevent model disagreement from being interpreted as an automatic correction of expert labels. First, the *reference annotation* re-

mains the authoritative benchmark record. Second, the *audit-priority score* serves as a transparent triage heuristic rather than an estimate of error probability. Third, *operational relevance* and *reviewer confidence* remain explicit human judgments and are not inferred from the technical score.

This paper makes four contributions:

1. We define a controlled protocol for evaluating zero-shot semantic naming by VLMs in an open-set tattoo scenario using original images and GT-derived oracle crops;
2. We provide a comparative analysis of three open-weight VLMs, focusing on semantic accuracy, *unknown* behavior, overprediction, omission, and cross-background stability;
3. We present *TattooAudit*, a functional benchmark-grounded interactive system that integrates review-queue triage, visual and model evidence, transparent priority indicators, structured expert actions, decision history, and exportable records; and
4. We delimit the technical and operational claims of the study by showing how benchmark evidence can support expert review and dataset-quality feedback without claiming end-to-end forensic automation, demographic fairness, cultural interpretation, or validated operational usability.

2 Related Work

This work lies at the intersection of tattoo analysis, open-set VLM evaluation, critical perspectives on biometric use, and interactive human oversight.

2.1 Tattoo datasets, detection, and semantic analysis

Tattoo research has traditionally focused on retrieval and matching, particularly when tattoos serve as soft biometric cues. The NIST Tatt-C challenge established an important public benchmark [Ngan and Grother, 2015]. Subsequent studies introduced deep-learning retrieval methods [Di and Patel, 2016], and TattTRN later revisited the task through stronger representation learning and semi-synthetic data [Gonzalez-Soler *et al.*, 2024]. Other work has investigated tattoo detection and classification through transfer learning [da Silva and Lopes, 2021, 2022], privacy-sensitive detection pipelines [Hrkac *et al.*, 2016], and synthetic or semi-synthetic data generation [Gonzalez-Soler *et al.*, 2023]. Most of these studies assume closed-set or otherwise controlled recognition settings.

Brilhador *et al.* extended this line of research by introducing TSSD2023 and formulating tattoo understanding as an open-set semantic-segmentation problem [Brilhador *et al.*, 2024]. Their benchmark supports both closed-set and open-set evaluation and assigns UUC content to an *unknown* label during testing. We use TSSD2023 as the source of images, masks, and semantic references, but shift the analytical focus from pixel prediction to semantic naming under oracle-segmentation conditions.

2.2 Open-set recognition and VLM behavior under reduced context

Open-set recognition must distinguish among known categories while retaining the ability to reject unfamiliar inputs

[Nunes et al., 2023]. Research on open-vocabulary segmentation shows that visual–semantic alignment remains fragile under domain shift, limited context, and noisy backgrounds [Liu et al., 2024b; Xie et al., 2024]. Tattoo imagery intensifies these challenges because individual elements may be highly stylized, partially occluded, or meaningful only within a larger composition [Brilhador et al., 2024].

VLMs are attractive because they generate flexible natural-language outputs without requiring task-specific training for every vocabulary [Radford et al., 2021]. However, they may also produce prompt-sensitive and semantically plausible hallucinations, particularly when the visual evidence is incomplete [Liu et al., 2024a; Bai et al., 2024; Seth et al., 2024]. Aggregate accuracy alone therefore cannot establish reliability [Kostumov et al., 2024]. We use the original-image and GT-crop conditions as controlled probes: when a model succeeds only with global context, its prediction may reflect contextual association rather than robust recognition of the isolated tattoo element.

2.3 Critical perspectives on tattoo biometrics and cultural taxonomies

Tattoo-based biometric systems are not socially neutral. Tattoos may reveal personal, religious, political, cultural, or community affiliations that would otherwise remain private. Bacchini and Lorusso argue that tattoo-recognition and tattoo-similarity technologies may subject tattooed individuals to disproportionate monitoring and discriminatory profiling, particularly when law-enforcement databases overrepresent specific demographic groups. They also caution against inferring attitudes, behavior, or group affiliation from visual similarity alone [Bacchini and Lorusso, 2018].

An object-level vocabulary supports reproducible evaluation, but it cannot capture every culturally situated or composite meaning. A traditional, ritual, memorial, or community-specific design may function as a semantic whole rather than as a set of independent Western object categories. TSSD2023 does not provide validated metadata on skin tone, ethnicity, cultural origin, social meaning, or institutional use. We therefore do not evaluate demographic disparities, infer cultural meaning, or claim that the benchmark taxonomy is culturally complete. Instead, we treat these omissions as explicit limits on our claims and as requirements for future datasets and governance protocols.

2.4 Interactive expert review and public-interest systems

Human-facing AI systems should support active inspection rather than present model outputs as self-sufficient evidence. Lopes et al. [2022] analyzed machine-learning interfaces through interactive-machine-learning principles, whereas Delcourt et al. [2024] emphasized user-centered and context-aware interaction. Research on model cards likewise demonstrates the value of structured documentation while cautioning that documentation alone cannot resolve questions of trust or appropriate use [Nunes et al., 2024].

Research in public-facing domains reinforces the need to connect automated detection with human interpretation and action. Accessibility assessment benefits from combining automated checks, specialist inspection, and user evaluation

[Mateus et al., 2021]. Studies of government chatbots show that usability and trust shape adoption [Monteiro et al., 2023]. Other work has examined how interactive systems can move from problem detection to actionable intervention [Palomino et al., 2025] and support human scrutiny of irregularities in public procurement [Braz et al., 2024]. TattooAudit follows this line of reasoning by converting benchmark discrepancies into a review workflow in which the expert remains responsible for the final action and its rationale.

The reviewed literature includes public-interest and human-centered AI research from institutions in developing countries, but direct studies of forensic human–AI interaction from underrepresented regions remain limited. We treat this limitation as a research gap, not as evidence that the problem is absent. The present artifact therefore offers a bounded technical foundation for future studies involving regionally situated experts, explicit governance requirements, and culturally appropriate evaluation protocols.

3 Study Design

We organize the study into three layers: (i) controlled VLM evaluation; (ii) deterministic generation of case-level prioritization indicators; and (iii) functional validation of the interactive research artifact. We did not recruit human participants. Accordingly, the study evaluates model behavior and artifact functionality, but it does not establish usability, trust calibration, decision quality, or operational effectiveness with domain experts.

3.1 Dataset and task scope

We use the publicly available TSSD2023 benchmark [Brilhador et al., 2024]. The dataset contains 2,106 images: 1,404 training images, 387 validation images, and 254 images in the public closed-set test split. An additional 61 images containing UUC content expand the public open-set split to 315 images. The benchmark defines 33 KKC semantic classes and 23 UUC semantic types; during evaluation, all UUC pixels share the *unknown* label [Brilhador et al., 2024].

The original task evaluates pixel-level semantic segmentation. In our protocol, we instead treat the reference masks as oracle segmentations. TSSD2023 supplies the original images, pixel-level reference masks, and reference labels used in this study. The preprocessing stage creates one crop for each non-background semantic class present in an image, rather than one crop for the image as a whole. For each class, it discards masks smaller than 200 pixels, computes a bounding box around all pixels assigned to that class, adds six pixels of padding while remaining inside the image boundary, and replaces pixels outside the class mask with either black (0) or white (255). This procedure produced 519 class crops from the 254 *test_closed* images and 632 class crops from the 315 *test_open* images for each background condition.

The resulting crops are derived experimental artifacts; they are not additional source images supplied by a future operator. Each crop is processed independently by a VLM. For an image containing multiple reference classes, the evaluation pipeline forms the image-level prediction by taking the set union of the normalized labels returned for its class-specific crops and removing duplicates. The image-level reference set is the union of the non-background class labels present in

the corresponding reference mask. The VLMs do not predict masks. This design holds segmentation quality constant and isolates the semantic-naming stage.

The quantitative comparison covers both public test splits. The interactive system uses a fixed *test_open* review queue because this split contains the unknown-content cases and provides the most informative setting for open-set expert review.

3.2 Input conditions, models, and execution environment

We evaluate every case under three visual conditions:

- **Original image:** the unmodified full image;
- **GT-crop black:** the reference-mask foreground rendered on a uniform black background; and
- **GT-crop white:** the same foreground rendered on a uniform white background.

The paired crops preserve the annotated foreground while changing a visually irrelevant background factor. This transformation partially controls visual context outside the annotated tattoo region, including surrounding skin appearance, illumination, and background texture. It therefore allows us to test whether semantic predictions remain stable when external contextual cues change. We do not interpret this protocol as a demographic-fairness evaluation. TSSD2023 does not provide validated metadata on skin tone, race, gender, geographic origin, or cultural affiliation, so we make no claims about performance differences across population groups. Such an investigation would require a purpose-built dataset, annotation protocol, and stratified evaluation.

All evaluations are zero-shot: none of the three VLMs is fine-tuned on TSSD2023. The models receive the same semantic task and the same 35-item output vocabulary, which excludes *background* and includes *unknown*: anchor, bird, branch, butterfly, cat, crown, diamond, dog, eagle, fire, fish, flower, fox, gun, heart, key, knife, leaf, lion, mermaid, octopus, owl, ribbon, rope, scorpion, shark, shield, skull, snake, spider, star, tiger, unknown, water, wolf. The execution scripts use model-specific serialization wrappers rather than literally identical prompt strings.

For Gemma3, the prompt requests a valid JSON object with the exact keys `labels` and `unknown`. The `labels` field must contain between zero and ten lowercase vocabulary items; the instructions prohibit generic terms, prohibit *background*, direct the model not to guess, and require `unknown=true` when the content falls outside the vocabulary or cannot be named confidently. For Qwen2.5-VL-7B, the prompt requests only a comma-separated list drawn from the same vocabulary, requires exact spelling, instructs the model to return *unknown* when uncertain or when no label applies, and prohibits any additional text or punctuation other than commas.

The LLaMA launcher preserves its literal prompt template. After inserting the controlled vocabulary, the user message is: “Return ONLY a comma-separated list of labels from this vocabulary: [VOCABULARY]. Rules: Use ONLY labels

from the list (exact spelling); If uncertain or none apply, return: unknown; No extra words, no punctuation other than commas.” The script passes the image and this text through the model’s chat template. The complete instantiated prompt strings and the shared vocabulary file are included with the experiment scripts.

Table 1 reports settings verified in the archived launchers and run metadata. Gemma3 was invoked as `gemma3:12b` through Ollama and ChatOllama, with temperature explicitly set to 0.0. Qwen used the checkpoint `Qwen/Qwen2.5-VL-7B-Instruct` through the Hugging Face `image-text-to-text` pipeline in `bfloat16`, with `max_new_tokens=64`; its scripts did not override other sampling arguments. LLaMA used `meta-llama/Llama-3.2-11B-Vision-Instruct` through Hugging Face Transformers, with `AutoProcessor` and `MllamaForConditionalGeneration`. The launcher sets `device_map=auto`, uses `bfloat16` when CUDA is available and `float32` otherwise, switches the model to evaluation mode, and executes inference under `torch.inference_mode()`. Original images and both crop conditions use the same generation call with `max_new_tokens=64` and `do_sample=False`. Thus, LLaMA uses greedy decoding; temperature and top-*p* are not passed to generation, and no random seed is set. CUDA synchronization immediately before and after generation makes the recorded wall-clock interval correspond to the completed GPU inference call.

We ran the experiments on the LABIC cluster at UTFPR with an Intel Xeon Gold 6338 CPU, one NVIDIA L40S GPU with 46 GB of VRAM, Debian 12, and host CUDA 12.2. The supplied Conda specification records Python 3.11, PyTorch, Torchvision, a PyTorch CUDA 12.1 package, Ollama, LangChain, `langchain-ollama`, Streamlit, and Pillow; not all pip package versions were pinned.

3.3 Prediction normalization and semantic metrics

The models return textual outputs through different serialization formats. The parser lowercases candidate labels, strips surrounding whitespace and delimiters, discards tokens outside the controlled vocabulary, removes *background*, and deduplicates the remaining labels. If a comma-list response contains no valid vocabulary item, the parser substitutes *unknown*. For JSON responses, the evaluation reads the `labels` list and applies the same vocabulary filter. The normalized *unknown* token remains an evaluable label: it can contribute to TP when the reference contains *unknown*, to FP when the reference does not contain it, or to FN when the reference contains it but the prediction omits it. The separately returned JSON boolean is retained as raw provenance but does not create an additional semantic label.

We represent each normalized image-level prediction as a set *Pred* and the reference annotation as a set *GT*. For crop conditions, *Pred* is the union of all crop-level prediction sets associated with the original image. We then compute:

$$\begin{aligned} TP &= |GT \cap Pred|, \\ FP &= |Pred \setminus GT|, \\ FN &= |GT \setminus Pred|. \end{aligned} \tag{1}$$

Table 1. Reproducibility configuration of the evaluated VLMs. The table reports only parameters verified in the archived launchers and run metadata.

Model	Checkpoint and runtime	Output and decoding configuration
Gemma3:12B	gemma3:12b; Ollama through ChatOllama.	JSON object with <code>labels</code> and <code>unknown</code> ; temperature 0.0. The launcher does not explicitly set <code>top-p</code> , a random seed, or an output-token limit.
Qwen2.5-VL-7B	Qwen/Qwen2.5-VL-7B-Instruct; Hugging Face image-text-to-text pipeline; bfloat16; CUDA device 0.	Comma-separated vocabulary labels; <code>max_new_tokens</code> =64. Other sampling arguments are not overridden by the launcher.
LLaMA 3.2 Vision 11B	meta-llama/Llama-3.2-11B-Vision-Instruct; Hugging Face AutoProcessor and MllamaForConditionalGeneration; <code>device_map</code> =auto; bfloat16 on CUDA and float32 otherwise.	Comma-separated vocabulary labels; <code>max_new_tokens</code> =64; <code>do_sample</code> =False (greedy decoding). Temperature, <code>top-p</code> , and a random seed are not passed.

In Equation 1, $\cap$ denotes set intersection, whereas $\setminus$ denotes set difference. Thus, $Pred \setminus GT$ contains labels predicted by the model but absent from the reference set, while $GT \setminus Pred$ contains reference labels that the model failed to recover.

These counts distinguish semantic matches, overprediction, and omission. We report precision, recall, micro-F1, and the Jaccard index as standard set-based and multi-label measures [Sokolova and Lapalme, 2009; Tsoumakas et al., 2010; Zhang and Zhou, 2014]. For two label sets A and B , the Jaccard index is

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}.$$

It ranges from 0, when the sets share no labels, to 1, when they are identical. We also report exact-set agreement (*perfect*), the proportion of normalized image-level predictions that contain *unknown*, and the average numbers of false positives and false negatives per image. The reported F1 evaluates semantic label sets; it is neither Dice nor mIoU because the VLMs do not predict masks.

For each split and condition, the pipeline sums TP, FP, and FN over images and computes micro-precision, micro-recall, and micro-F1 from these pooled counts. Table 6 pools the two public test splits by summing their counts over 569 original images and recomputing micro-F1; it does not average the two split-level F1 values. Runtime measures the wall-clock interval around each model call and excludes crop generation and metric computation. Original-image time is total original-image inference time divided by 569. Crop-condition time is the sum of all 1,151 class-crop inference times (519 closed-set and 632 open-set crops) divided by 569 original images; it therefore represents the accumulated crop-inference cost per original image rather than time per individual crop. The consolidated black/white Jaccard is the arithmetic mean of the two split-level pairwise-Jaccard values.

3.4 Cross-background stability

Using the set-similarity definition above, let $Pred_i^{black}$ and $Pred_i^{white}$ denote the label sets for the paired crops of case i , and let N denote the number of paired cases. We compute the average pairwise Jaccard index:

$$J_{b/w} = \frac{1}{N} \sum_{i=1}^N \frac{|Pred_i^{black} \cap Pred_i^{white}|}{|Pred_i^{black} \cup Pred_i^{white}|}. \quad (2)$$

We also compute a descriptive black/white flip rate:

$$\text{FlipRate}_{b/w} = \frac{100}{N} \sum_{i=1}^N I(Pred_i^{black} \neq Pred_i^{white}), \quad (3)$$

where $I(\cdot)$ denotes the indicator function. Higher $J_{b/w}$ values and lower flip rates indicate greater stability under background variation.

3.5 System data pipeline and case construction

The complete study pipeline separates offline artifact generation from interactive review. During the offline stage, our preprocessing derives the black- and white-background crops from the benchmark masks, invokes each VLM on the original image and both crops, normalizes the responses, and computes case-level metrics. TattooAudit begins after this stage. It consumes the persisted artifacts rather than generating masks or crops or invoking the VLMs during review. Within the interactive stage, the implementation constructs each reviewable case through the following deterministic pipeline:

1. **Case discovery:** the system identifies case identifiers from the GT-crop directory or the prepared panel-assets directory for *test_open*.
2. **Artifact loading:** for each model and visual condition, it loads per-image metrics, per-crop metrics, predicted-label tables, and best/worst-case tables from the corresponding evaluation directory. The loader filters records by split and removes duplicated per-image rows.
3. **Fallback recovery:** when a normalized prediction is unavailable in a CSV table, the system searches the corresponding raw JSONL output and extracts the parsed labels and *unknown* field.
4. **Reference recovery:** the system obtains the current reference labels from the GT-crop filenames when the evaluation tables do not provide them.
5. **Metric completion:** when FP or FN values are missing, the system reconstructs them from normalized prediction and reference sets.
6. **Priority construction:** the system applies the transparent rules in Equation 4, sorts cases by score, number of indicators, and identifier, and then joins the latest human-review metadata.
7. **Interactive review:** the interface aligns the selected case with its original image, reference mask, crop pairs, model evidence, priority reasons, and prior decisions.

8. **Record preservation:** each submitted action is added to the history-preserving expert-review table, from which the interface derives the latest status while retaining earlier decisions.

Separating offline model execution from online review makes the artifact reproducible and responsive. It also bounds our claim: TattooAudit validates an expert-review workflow over a fixed benchmark export rather than an end-to-end operational inference service.

An operational extension could begin with a repository that contains only original images, but it would require an explicit upstream image-preparation stage. When trusted masks are available, the pipeline can derive the black- and white-background crops deterministically. When masks are unavailable, the workflow must first obtain them through qualified manual annotation or a separately validated segmentation model. We do not implement or evaluate this extension; therefore, the present study does not claim end-to-end processing of unseen operational images.

3.6 Evaluation layers and claim boundaries

Figure 1 summarizes the complete study and makes the system boundary explicit. The offline preprocessing and evaluation layer starts from benchmark original images, reference masks, and reference labels; derives the two crop conditions; executes the VLMs; and produces normalized predictions and quantitative metrics. The prioritization layer converts selected case-level patterns into transparent indicators. The interactive layer consumes these precomputed artifacts, presents the evidence, records an expert action, and preserves the decision history. Each layer answers a distinct question, and the three should not be conflated.

For RQ1, we use the quantitative metrics defined above. For RQ2, we verify whether the implemented artifact can load all 315 *test_open* cases, reproduce the deterministic priority ranking, align each case with its visual and model evidence, collect structured decisions, retain multiple decisions in the history, derive the latest review status, and export both the queue and the review log. This procedure constitutes functional artifact validation, not a human-participant usability evaluation.

4 TattooAudit: Interactive Expert-Review System

TattooAudit is a benchmark-grounded review workspace for specialists who understand the tattoo-label vocabulary and can assess annotation quality, ambiguity, and routing needs. It operates downstream of the offline preprocessing and evaluation pipeline. The current interface does not classify new operational images end to end, generate reference masks, create the two crop conditions, or invoke the VLMs. Instead, it consumes precomputed original-image and crop predictions, reference masks, GT-derived crops, and case-level metrics.

4.1 Design requirements

We derived six design requirements from the study goals and from the need to prevent VLM outputs from being presented

as final authority. Table 2 maps each requirement to its implemented mechanism.

4.2 Information architecture and review workflow

The interface combines a persistent sidebar with four task-oriented tabs. The sidebar contains a minimum-priority slider, a review-status filter, a case selector, and pending/reviewed counters. The system fixes the dataset to *test_open*; we removed the earlier split selector to align the interactive artifact with the article’s open-set focus. The case selector displays each case identifier and priority value, and selecting a case immediately opens the corresponding visual evidence in the main workspace. The current implementation does not yet support thumbnail-based browsing within the selector. We retain this feature as a usability limitation and as a target for the future user-study version.

The four tabs implement a progressive review workflow:

1. **Case review:** inspect the original image, reference mask, current reference labels, optional GT-derived crops, prioritization reasons, latest decision, and expert-decision form;
2. **Model evidence:** compare the three models under original-image, black-crop, and white-crop conditions, including predicted labels, FP, FN, *unknown*, context-stability assessment, and expandable technical metrics;
3. **Decision history:** inspect all saved decisions for the selected case and export the complete review log; and
4. **Review queue:** inspect and export the global case list enriched with priority, latest action, reviewer, and last-review timestamp.

4.3 Transparent audit-priority mechanism

The prioritization mechanism translates selected model behaviors into a deterministic case-level index. It does not estimate the probability that the reference annotation is incorrect. For each model m , the system computes four indicators: U_m for the presence of *unknown* in any condition, F_m for a black/white flip, D_m for original-image/crop divergence, and $H_{m,v}$ for high overprediction ($FP \geq 3$) in condition v . For case c :

$$P(c) = \sum_{m \in M} \left[2U_m(c) + 2F_m(c) + D_m(c) + 2 \sum_{v \in V} H_{m,v}(c) \right], \quad (4)$$

where M contains the three evaluated models and $V = \{\text{original}, \text{black}, \text{white}\}$. Each model can contribute at most 11 points, which yields a theoretical maximum of 33. The highest score observed in the current *test_open* export is 27. The queue sorts cases by descending score, descending indicator count, and case identifier.

Table 3 states the scoring rules and qualitative bands explicitly. A score of zero means that none of the current rules was triggered; it does *not* imply exact agreement with the ground truth. For example, an isolated false positive below the high-FP threshold may coexist with a score of zero. The sidebar’s minimum-priority slider only filters the queue. It

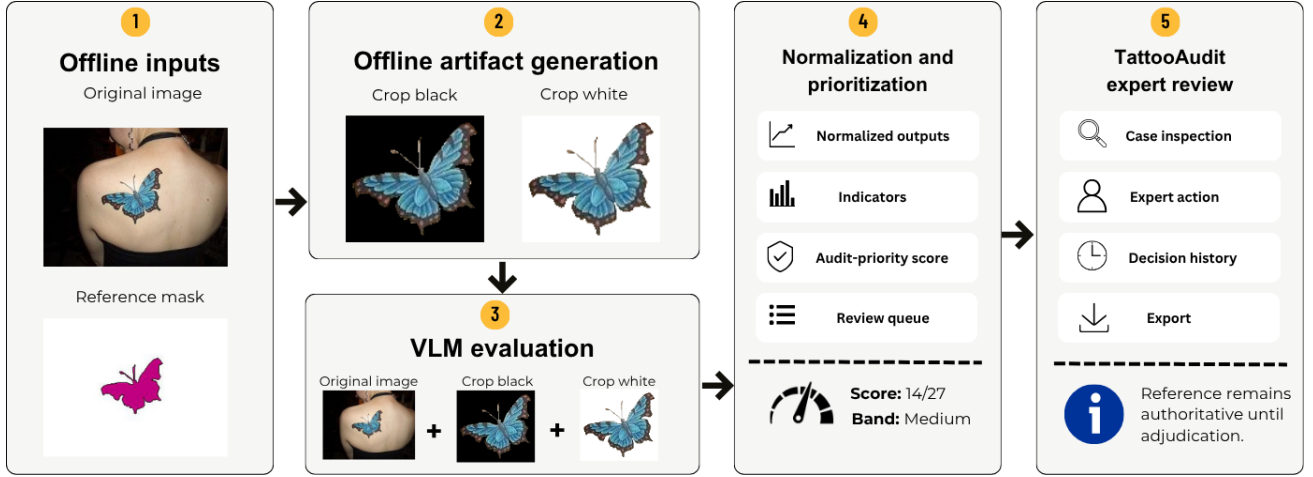


Figure 1. Study workflow and system boundary. The offline preprocessing and evaluation stage starts from TSSD2023 original images, reference masks, and reference labels; derives crops on black and white backgrounds; executes the three VLMs on the original image and both crops; and stores normalized outputs and case-level metrics. The prioritization layer applies deterministic rules and constructs the review queue. TattooAudit consumes these precomputed artifacts to support case selection, evidence inspection, expert action, decision history, and export; it does not generate masks, crops, or VLM predictions during the interactive session. The reference annotation remains authoritative until an explicit adjudication process approves a revision.

Table 2. Design requirements and their implementation in TattooAudit.

Requirement	Rationale	Implemented mechanism
Transparent triage	The user must know why a case appears early in the queue.	Relative audit-priority score, qualitative band, indicator count, and human-readable reasons.
Evidence before action	A decision should follow inspection of the reference and model evidence.	Original image, reference mask, current reference labels, GT-derived crops, and cross-model evidence table.
Human authority	Model disagreement must not automatically overwrite the benchmark.	Reference annotation is explicitly identified as primary; revisions are stored as proposals in a separate log.
Actionable review	Inspection should lead to a defined next step.	Confirm, revise, mark ambiguous, record a re-annotation request, or record an exclusion recommendation for the high-confidence pool.
Traceability	Later reviewers must reconstruct who decided what and why.	Reviewer identifier, timestamp, confidence, operational relevance, rationale, decision history, and CSV export.
Separation of signals	Technical instability and operational importance are distinct constructs.	The system computes technical priority; the reviewer separately records confidence and operational relevance.

does not change Equation 4, the observed maximum, or the stored score.

Table 3. Audit-priority indicators and qualitative bands.

Indicator	Trigger	Weight
Unknown present	<i>unknown</i> appears in any condition for a model	+2
Black/white flip	Crop label sets differ across backgrounds	+2
Original-image/crops divergence	Original-image prediction differs from either crop result	+1
High FP	$FP \geq 3$ in one condition (original image(+2), black crop(+2), white crop(+2))	max. +6 (+2 per cond.)

Bands: 0–3 Very low; 4–9 Low; 10–16 Medium; 17–20 High; ≥ 21 Critical.

The present prototype uses technical instability only to rank cases; it does not implement a mandatory human-review floor based on sensitive content or institutional policy. All cases remain accessible in the queue, including those with low scores. An operational version should add an independent governance rule $G(c)$ so that a case may require review even when its technical score is low:

$$\text{ReviewRequired}(c) = I(P(c) \geq \tau) \vee G(c), \quad (5)$$

where $G(c)$ must derive from authorized metadata, an explicit expert flag, or a documented institutional policy—not

from an unsupported VLM inference about cultural, criminal, or social meaning. Equation 5 defines a future governance requirement and is not implemented in the current code.

4.4 Case evidence and model evidence

The *Case review* tab presents reference evidence before model evidence. It displays the original image and reference mask side by side, followed by the current expert reference. An expandable panel reveals each GT-derived black/white crop pair. The interface then lists, in plain language, the exact indicators that produced the case score.

The *Model evidence* tab uses a compact table with one row for each model–condition pair. The table reports predicted labels, FP, FN, the presence of *unknown*, and a model-level context assessment (*stable across contexts* or *context-sensitive*). Collapsible model panels preserve access to detailed metrics without forcing the primary screen to display every field at once.

4.5 Structured expert action and feedback loop

After inspecting the evidence, the reviewer selects one of five structured actions:

- **Confirm current reference:** retain the current reference labels;
- **Propose revised reference:** record a candidate label set without modifying the original benchmark files;

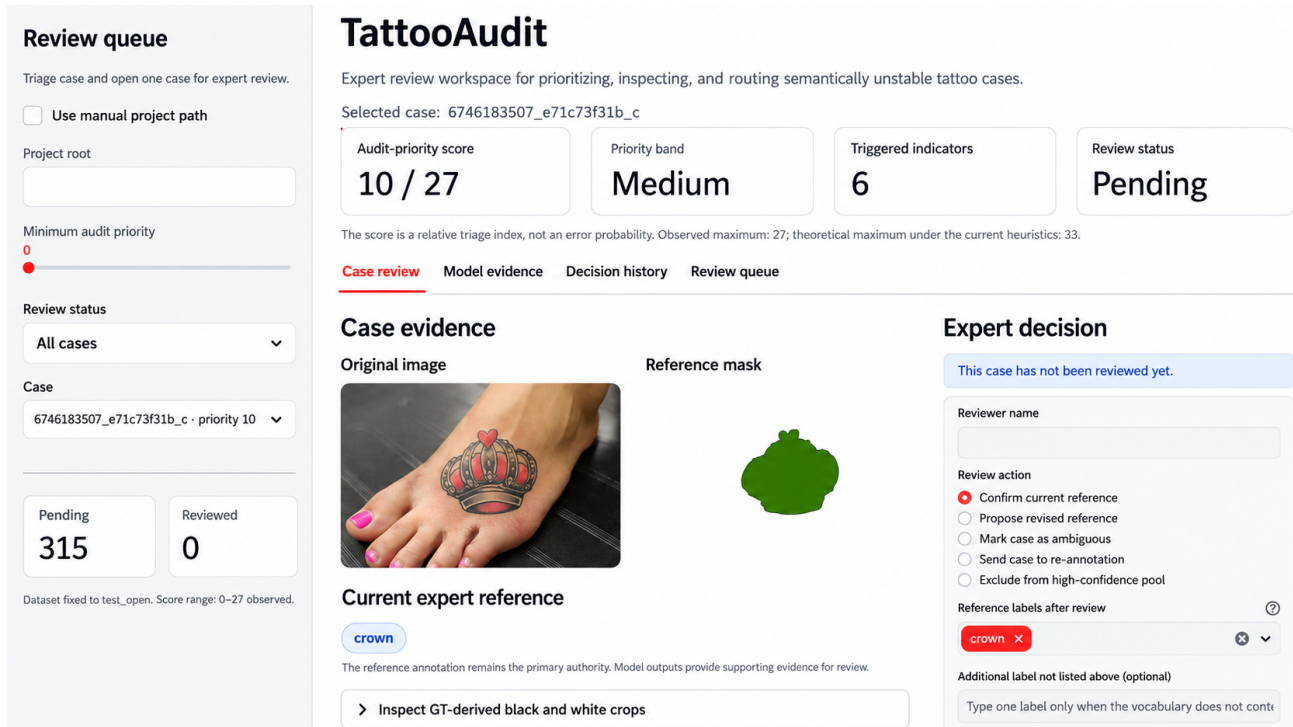


Figure 2. Screenshot of the TattooAudit overview. The sidebar filters the fixed *test_open* review queue by minimum priority and review status. The header exposes the selected case, relative score, qualitative band, indicator count, and latest review status. The tabs separate case evidence, model evidence, decision history, and the global review queue.

- **Mark case as ambiguous:** state that the available visual evidence does not support a sufficiently clear interpretation;
- **Send case to re-annotation:** record a routing request for a future annotation or consensus process; or
- **Exclude from high-confidence pool:** record that the case should not be treated as part of a high-confidence reference subset.

The form also records the reviewer’s name, the labels after review, an optional out-of-vocabulary label, reviewer confidence, operational relevance, and a rationale. The implementation requires a reviewer identifier for every saved decision. It also requires a rationale for revision, ambiguity, routing, and exclusion actions, as well as at least one label when the reviewer proposes a revised reference.

Table 4 summarizes the stored schema. The system writes decisions to a separate `expert_review_log.csv`, which supports a feedback loop without silently modifying the benchmark. To accommodate schema evolution during prototyping, the implementation reads the existing table, appends the new row, and rewrites the complete CSV while preserving all earlier records. The most recent timestamped decision determines the displayed review status, whereas the history tab retains the full sequence of prior decisions.

4.6 Operational-relevance field and governance boundary

The operational-relevance field contains reviewer-entered metadata. The model does not predict this field. The field neither alters the technical priority nor automatically triggers an external action. Table 5 defines its intended interpretation in the research prototype. An operational deployment

would require an institutionally approved policy that specifies qualified reviewers, evidence requirements, escalation paths, access control, retention, and adjudication procedures.

Table 5. Interpretive guidance for the operational-relevance field.

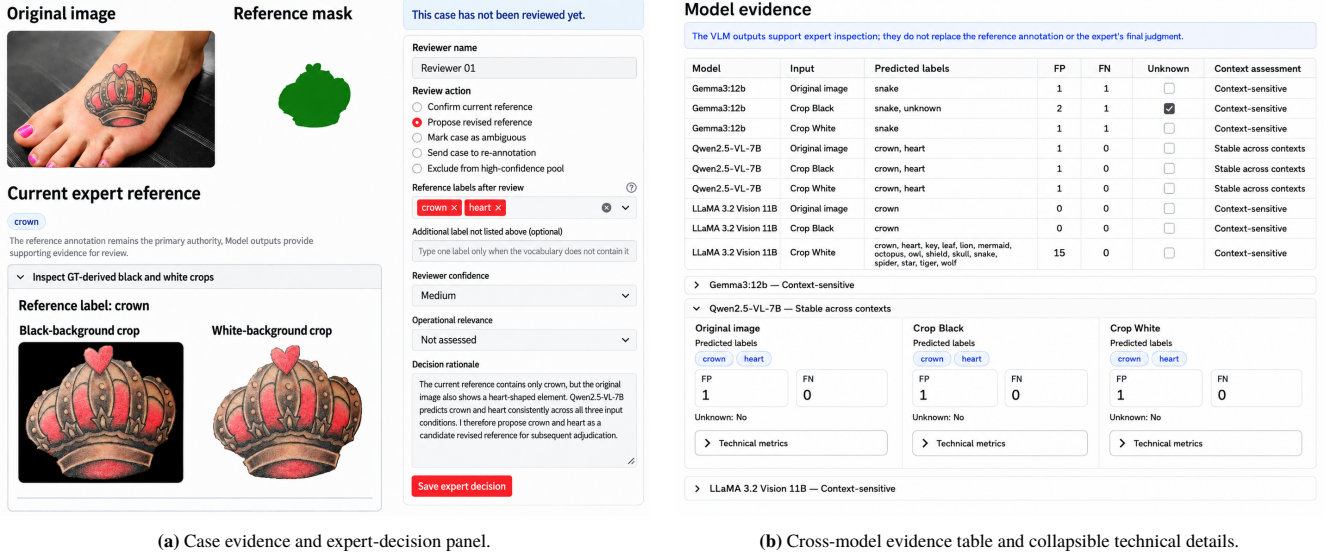
Value	Intended interpretation in the prototype
Not assessed	The reviewer lacks sufficient contextual information or chooses not to make an operational judgment.
Low	The issue is primarily a benchmark-quality or documentation concern.
Medium	The issue may affect comparative analysis, case reuse, or a subsequent annotation decision.
High	The reviewer believes the case may require secondary review under a future governance protocol because an error could carry a sensitive consequence.

The prototype stores this field for traceability. We do not claim that the categories have been validated or that a “High” value constitutes an official public-policy or forensic determination.

The global review queue combines the deterministic ranking with the latest human-review metadata. It displays the case identifier, audit priority, qualitative band, number of indicators, review status, latest action, reviewer, and most recent review timestamp. Users can export both the review queue and the complete review log as CSV files.

5 Results

We first report the quantitative model comparison and then show how TattooAudit transforms the benchmark results into case-level review evidence.



(a) Case evidence and expert-decision panel.

(b) Cross-model evidence table and collapsible technical details.

Figure 3. Screenshots of the evidence-centered review. The interface presents the reference annotation as primary evidence and uses VLM outputs as supporting information for inspection rather than as automatic corrections.

Table 4. Structured expert-review schema.

Field	Role
reviewer, timestamp	Identify who recorded the decision and when.
review_action, review_status	Store the selected action and its derived workflow status.
original_reference_labels	Preserve the benchmark reference visible at review time.
reviewed_reference_labels	Store the confirmed or proposed post-review label set.
reviewer_confidence	Record High, Medium, Low, or Not assessed.
operational_relevance	Record a separate human assessment of Low, Medium, High, or Not assessed.
priority_score, priority_band, priority_flags	Preserve the technical context that motivated inspection.
rationale	Preserve the reviewer’s evidence-based explanation.

5.1 Comparative evaluation of model performance

Table 6 summarizes the two public test splits using the pooling and runtime-accounting procedure defined in Section 3.3. Qwen2.5-VL-7B achieves the strongest overall balance of semantic accuracy, runtime, and black/white stability. Gemma3 degrades more substantially under crop conditions and frequently produces *unknown*. LLaMA 3.2 Vision 11B performs moderately on original images but generates many false positives under crop conditions.

Figures 5 and 6 show the main trend on *test_open*. Qwen2.5-VL-7B maintains the strongest performance across the original-image condition and both crop conditions. Gemma3 and LLaMA decline more sharply when the task removes global context.

Table 7 reports the detailed *test_open* results. Qwen2.5-VL-7B reaches a micro-F1 of 0.728 on the original image and remains comparatively strong on the black and white crops (0.644 and 0.678, respectively). In Gemma3’s crop conditions, *unknown* usage exceeds 65%, and F1 decreases. LLaMA exhibits a different failure pattern: crop-based FN remains relatively low, but FP rises sharply, indicating overprediction rather than reliable semantic recovery.

For the black/white crop pairs, Qwen2.5-VL-7B reaches Jaccard values of 0.852 on *test_open* and 0.849 on *test_closed*, with flip rates of 27.1% and 28.3%, respectively. Gemma3 remains near 0.69 Jaccard and produces flip rates above 63%, whereas LLaMA remains near 0.63 Jaccard with flip rates of

approximately 57%. Because the paired crops preserve the foreground, these differences reveal sensitivity to the background rendering.

Figures 7 and 8 clarify the model-specific failure patterns. LLaMA’s crop behavior is dominated by FP inflation. Gemma combines omission, weak crop performance, and frequent use of *unknown*. Qwen remains more balanced and substantially more stable across the two backgrounds, although it also uses *unknown* frequently under the open-set protocol.

5.2 Interpreting aggregate errors and the *unknown* token

The aggregate metrics characterize model behavior but do not identify a single cause for each failure. An omitted or additional label may result from degraded visual evidence, stylization, overlap, a composite motif, a limitation of the reference taxonomy, an out-of-vocabulary concept, or a model error. Similarly, the normalized *unknown* token indicates rejection or unresolved naming within the benchmark protocol, but it does not reveal which cause applies. TattooAudit therefore exposes the original evidence, crops, reference labels, model outputs, and technical metrics so that the reviewer can document a case-level interpretation. The current study includes neither an adjudicated taxonomy of failure causes nor demographic strata or cultural metadata, and it does not use aggregate F1 to support fairness claims.

Decision history

timestamp	reviewer	review_status	review_action	original_reference_labels	reviewed_reference_labels	reviewer_confide	operati
2026-07-18T17:06:02-03:00	Reviewer 01	Revision proposed	Propose revised reference	crown	crown;heart	Medium	Not ass

[Download expert review log](#)

Review queue

The queue preserves the original audit-priority heuristic and adds the latest human-review status.

Case	Audit-priority	Band	Indicators	Review status	Latest action	Reviewer	Last reviewed
6710754759_8e13227a39_c	11 / 27	Medium	6	Pending			
21850693655_22796ef0d2_z	10 / 27	Medium	6	Pending			
2598214054_37a3b28f97_c	10 / 27	Medium	6	Pending			
3102166900_e7df9c3d16_c	10 / 27	Medium	6	Pending			
3546531666_fb96710689_b	10 / 27	Medium	6	Pending			
6746183507_e71c73f31b_c	10 / 27	Medium	6	Revision proposed	Propose revised refe	Reviewer 01	2026-07-18T17:06:02-0
8165350829_cc1a0ea333_z	10 / 27	Medium	6	Pending			
36845095386_4af709ef22_b	9 / 27	Low	6	Pending			
8165424962_7d52e46d84	9 / 27	Low	6	Pending			
136246728_ff20e95f22	9 / 27	Low	5	Pending			

[Download review queue](#)

Figure 4. Screenshot of the TattooAudit decision-history and review-queue views after saving an illustrative review record for functional validation of case 6746183507_e71c73f31b_c. The upper panel shows the persisted review record, including timestamp, reviewer identifier, review status, review action, original reference, reviewed reference labels, and confidence level. The lower panel shows the updated review queue, where the same case now appears with the status *Revision proposed*, the latest action, reviewer identifier, and last-reviewed timestamp. Together, these views illustrate the traceability of human review and the propagation of adjudication metadata to the global queue.

Table 6. Consolidated performance by model and condition across *test_open* and *test_closed*.

Model	F1 original	F1 black	F1 white	Time original (s/img)	Time black (s/img)	Time white (s/img)	Jaccard (b/w)
Gemma3:12B	0.307	0.200	0.204	1.244	4.128	3.891	0.692
Qwen2.5-VL-7B	0.725	0.639	0.668	0.726	1.258	0.890	0.850
LLaMA 3.2 Vision 11B	0.565	0.289	0.355	1.609	5.911	5.741	0.633

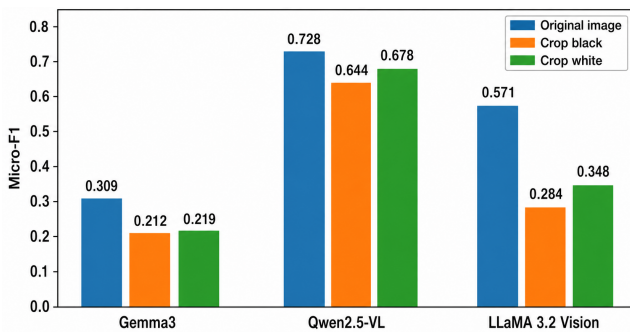


Figure 5. Micro-F1 by model and visual condition on *test_open*.

5.3 Priority distribution and case-level interpretability

The 315 *test_open* cases receive scores ranging from 0 to 27 (mean 16.06; median 17.00). Under the current bands, 7 cases fall into Very low, 50 into Low, 92 into Medium, 75 into High, and 91 into Critical. These values characterize the current heuristic and model set. They are not calibrated probabilities and should not be generalized to another collection of models

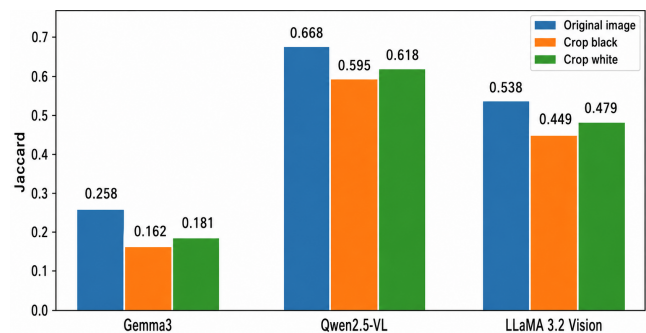
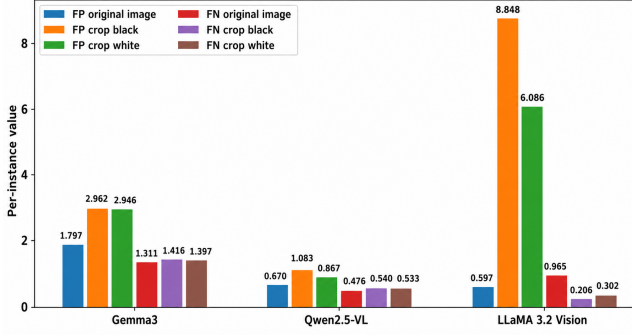
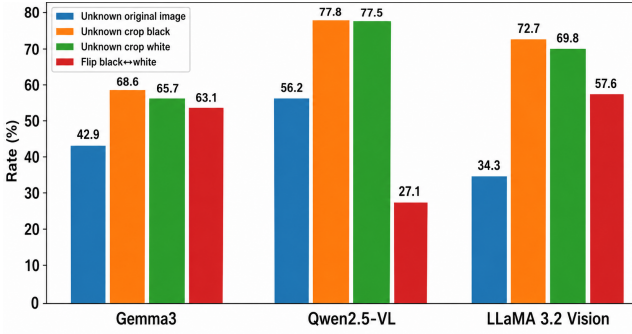


Figure 6. Jaccard by model and visual condition on *test_open*.

Table 7. Detailed semantic naming results on *test_open*.

Model	Condition	F1 (micro)	Jaccard	Perfect	Unknown	FP/img	FN/img
Gemma3:12B	original image	0.309	0.258	11.1%	42.9%	1.797	1.311
Gemma3:12B	crops black	0.212	0.162	3.2%	68.6%	2.962	1.416
Gemma3:12B	crops white	0.219	0.181	5.7%	65.7%	2.946	1.397
Qwen2.5-VL-7B	original image	0.728	0.668	38.4%	56.2%	0.670	0.476
Qwen2.5-VL-7B	crops black	0.644	0.595	30.2%	77.8%	1.083	0.540
Qwen2.5-VL-7B	crops white	0.678	0.618	32.1%	77.5%	0.867	0.533
LLaMA 3.2 Vision 11B	original image	0.571	0.538	32.7%	34.3%	0.597	0.965
LLaMA 3.2 Vision 11B	crops black	0.284	0.449	27.6%	72.7%	8.848	0.206
LLaMA 3.2 Vision 11B	crops white	0.348	0.479	27.0%	69.8%	6.086	0.302

**Figure 7.** False positives and false negatives per instance on *test_open*.**Figure 8.** Unknown rate and black/white flip rate on *test_open*.

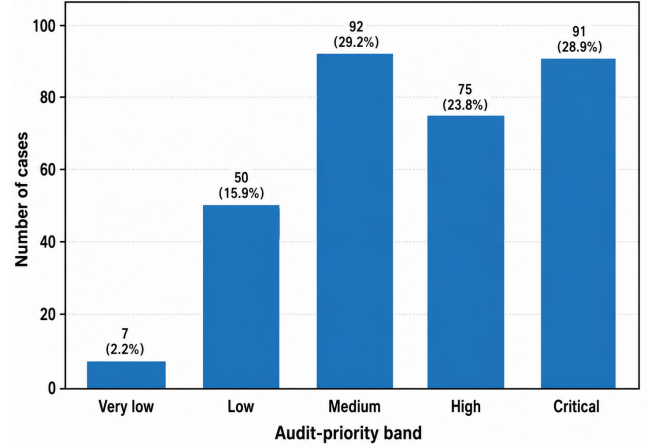
without recomputation.

The interface makes three recurring behaviors directly inspectable: (i) dependence on global context, when the original-image prediction and crop predictions differ; (ii) sensitivity to background rendering, when paired crops produce different labels; and (iii) model-specific failure patterns, such as *unknown* fallback or FP inflation. The score summarizes these behaviors, whereas the individual indicators preserve their explanation.

5.4 Illustrative expert-review walkthrough

The screenshots and stored record used in this walkthrough were created by the authors to verify the implemented interaction, persistence, and queue-update mechanisms. They do not represent an observation from a user study, an independently adjudicated expert decision, or evidence that the proposed revision is correct.

Table 8 summarizes case 6746183507_e71c73f31b_c, whose current benchmark reference contains only *crown*. The original image also contains a small heart-shaped element above the crown. TattooAudit assigns six indicators to the case and produces an audit-priority score of 10/27 (Medium). Gemma fails to recover the reference and predicts *snake* in all three conditions, while also adding *unknown* on the black

**Figure 9.** Distribution of audit-priority bands across the 315 *test_open* cases. Bars report the number and percentage of cases assigned to each qualitative band. The bands represent relative technical-instability priority under the deterministic heuristic; they do not indicate error probability or operational severity.

crop. Qwen consistently predicts both *crown* and *heart* for the original-image, black-crop, and white-crop conditions. LLaMA predicts *crown* correctly for the original image and black crop but strongly overpredicts on the white crop.

The visible indicators determine the score directly. Gemma contributes five points: *unknown* (+2), a black/white flip (+2), and original-image/crop divergence (+1). LLaMA also contributes five points through a black/white flip (+2), original-image/crop divergence (+1), and high overprediction on the white crop (+2). Qwen contributes no priority points because its prediction remains stable across conditions and triggers neither the *unknown* nor the high-FP rule. The total score is therefore 10.

This case illustrates a second role for TattooAudit beyond exposing model instability. Under the benchmark protocol, the system counts *heart* as a false positive because the reference contains only *crown*. However, the original image visibly contains a heart-shaped element, and Qwen identifies it consistently in all three input conditions. The interface therefore helps a specialist distinguish a probable model error from a plausible annotation omission. The specialist may then propose *crown*, *heart* as a revised reference or route the case for re-annotation. This example does not establish that the benchmark annotation is incorrect. Instead, it shows how case-level evidence can reveal a potentially incomplete reference that aggregate metrics would otherwise treat only as an error.

Table 8. Model evidence shown by TattooAudit for the illustrative *crown* case. The benchmark reference contains only *crown*; therefore, the evaluation counts Qwen’s visually plausible *heart* label as one false positive until expert review adjudicates the case.

Model	Condition	Predicted labels	FP	FN	Unknown
Gemma3:12B	Original image	snake	1	1	No
Gemma3:12B	Crop black	snake, unknown	2	1	Yes
Gemma3:12B	Crop white	snake	1	1	No
Qwen2.5-VL-7B	Original image	crown, heart	1	0	No
Qwen2.5-VL-7B	Crop black	crown, heart	1	0	No
Qwen2.5-VL-7B	Crop white	crown, heart	1	0	No
LLaMA 3.2 Vision 11B	Original image	crown	0	0	No
LLaMA 3.2 Vision 11B	Crop black	crown	0	0	No
LLaMA 3.2 Vision 11B	Crop white	crown, heart, key, leaf, lion, mermaid, octopus, owl, shield, skull, snake, spider, star, tiger, water, wolf	15	0	No

6 Discussion

6.1 From benchmark discrepancy to expert action

The revised system addresses a central limitation of the earlier dashboard design. A display that merely lists model outputs and a priority score may support inspection, but it does not define what the user should do next. TattooAudit connects four stages: triage, evidence inspection, expert action, and decision history. This interaction loop turns the system into an interactive decision-support artifact rather than a static visualization of benchmark results.

The review actions also redefine the human role. The reviewer does not merely confirm or reject the system’s priority. Instead, the reviewer evaluates the reference itself and selects an action related to dataset quality or workflow routing. The VLMs provide additional evidence, while the specialist retains authority to preserve the current reference, propose a revision, acknowledge ambiguity, request a new annotation process, or recommend removing the case from a high-confidence subset.

6.2 Design lessons for audit-oriented human-AI systems

The implementation yields four transferable design lessons. First, aggregate performance should not determine the review order by itself; case-level instability requires a separate and inspectable triage layer. Second, the interface should expose the reasons for prioritization rather than present a score without provenance. Third, model disagreement should initiate inspection rather than automatically replace a reference label. Fourth, a human decision becomes auditable only when the system records the action, evidence context, reviewer, timestamp, rationale, and prior history. These lessons extend beyond tattoos to other benchmark-grounded systems in which model outputs may reveal annotation problems but cannot establish ground truth independently.

6.3 Why transparent triage matters

The benchmark results confirm that context can improve VLM performance, but they also show that context may conceal fragility. A model may succeed on an original image yet fail on the corresponding crop, and identical foregrounds may produce different labels on black and white backgrounds. A transparent queue helps allocate limited review effort to cases that accumulate evidence of such instability.

The score remains intentionally simple and inspectable. Every point traces back to a visible rule and model condi-

tion. At the current stage, this transparency is preferable to an opaque learned-priority model. However, the score is not calibrated, and the selected weights express design priorities rather than empirically estimated costs. The interface therefore presents the score together with its qualitative band, indicator count, and textual reasons.

6.4 Technical priority is not a governance policy

A continuous technical ranking cannot, by itself, represent the asymmetric consequences of high-stakes errors. A low-scoring case may still require review because of external context, a prior expert concern, or an institutionally defined sensitive category. Conversely, a critical technical score indicates accumulated model instability rather than criminal relevance or social risk. The future governance rule in Equation 5 therefore remains independent of the benchmark score. This separation prevents a model-derived heuristic from deciding which culturally, legally, or socially sensitive cases deserve human attention.

6.5 Human authority, provenance, and responsible use

TattooAudit separates reference data, model evidence, and human feedback. A proposed revision does not overwrite the benchmark; instead, it becomes a new review record. This history-preserving design maintains provenance and supports later adjudication. The decision history also prevents the latest-state view from erasing disagreement among reviewers or changes over time.

The system also separates technical priority from operational relevance. A technically unstable case is not necessarily operationally important, and a low-priority case is not necessarily safe for deployment. The reviewer records operational relevance and confidence explicitly. This distinction is particularly important for tattoos because their interpretation may carry social and ethical implications [Bacchini and Lorusso, 2018].

In victim-identification or investigative-support scenarios, omission, unsupported association, and re-identification risk may produce different harm profiles [Fang et al., 2018]. The interface cannot resolve these harms by itself. Its responsible role is narrower: preserve the evidence that motivated review, prevent silent model-driven changes, and make the human decision traceable. Any operational use would require qualified personnel, legal authority, data-minimization rules, access controls, and a documented appeal or adjudication

process.

6.6 Review records as a future learning resource

The structured review table may support future model and dataset improvement because each row links the case identifier, original reference, proposed or confirmed labels, model-derived priority context, reviewer confidence, operational relevance, and rationale. After independent adjudication and quality control, these records could form a correction dataset for supervised fine-tuning, prompt refinement, error analysis, or evaluation of later model versions.

Raw review logs should not be used directly for training. A learning pipeline would need to remove or protect sensitive identifiers, reconcile conflicting reviewer decisions, retain provenance, separate training and evaluation cases, document the authority behind each correction, and examine whether the collected feedback reproduces cultural or institutional bias. The current artifact exports the records but does not perform fine-tuning or claim that a single reviewer decision constitutes a verified training label.

7 Limitations and Future Work

The study has several technical limitations. First, oracle segmentation removes mask-prediction errors and therefore does not represent an end-to-end operational pipeline. Second, the semantic vocabulary and output-normalization procedure constrain the VLMs’ open-ended responses. The supplied artifact preserves the execution scripts, prompt templates, normalized outputs, and timing summaries for the three evaluated models. Exact bitwise reproduction can still depend on hardware, driver, and library behavior because the experiments did not enforce deterministic CUDA algorithms or pin every pip dependency. Third, the audit-priority score relies on manually specified rules and weights. Although transparent, it is not calibrated against expert cost, review time, or confirmed annotation error. Fourth, the current artifact uses precomputed predictions and a fixed *test_open* queue rather than processing new images interactively. Fifth, the case selector does not yet support thumbnail-based browsing, and the interface has not been empirically optimized for high-volume inspection. Sixth, a future repository containing only original images would require an upstream segmentation or annotation component before the controlled black/white crop comparison could be reproduced.

The current priority mechanism also lacks an implemented mandatory-review governance floor. The operational-relevance and reviewer-confidence fields contain self-reported metadata; they do not change the queue, trigger escalation, or represent validated constructs. An operational version would require a policy layer independent of model uncertainty, including role-based access, escalation rules, secondary review, adjudication, and audit-log protection.

The benchmark is static and does not represent the temporal evolution of tattoos or skin. Fading, scarring, pigment diffusion, aging, partial covering, retouching, and image acquisition years apart may change both the visible evidence and the model response. A longitudinal study would require repeated images of the same tattoo, documented acquisition conditions, and an evaluation of semantic stability over time.

The reported aggregate metrics may conceal label-specific variation. We do not provide validated stratification by tattoo category, skin tone, demographic group, cultural tradition, or failure cause. The dataset lacks the metadata required for demographic and cultural analyses, and the current outputs do not distinguish whether *unknown*, FP, or FN results arise from image degradation, taxonomy limitations, composite imagery, or model behavior. Future work should report class-level performance and support counts, incorporate expert-coded failure causes, and use carefully governed demographic or cultural metadata where ethically and legally appropriate.

Most importantly, we have not conducted a user study with tattoo-domain or public-safety specialists. Functional artifact validation demonstrates that the workflow is implemented, but it does not establish usability, cognitive workload, trust calibration, inter-rater agreement, or improved review accuracy. A future evaluation should recruit qualified experts and compare at least three conditions: (i) review without the dashboard; (ii) review with aggregate model metrics only; and (iii) review with the complete TattooAudit workflow. Relevant outcomes include task-completion time, decision accuracy against adjudicated references, rationale quality, action consistency, perceived transparency, workload, and inter-rater reliability.

Future work should integrate learned segmentation, support multiple independent reviewers and adjudication, provide configurable scoring and governance policies, evaluate alternative thresholds, add thumbnail navigation and label-level provenance, and test whether adjudicated reviewer feedback can improve future dataset versions, prompts, or fine-tuned models. Any operational study involving sensitive records should follow an institutionally approved protocol and appropriate data-governance safeguards.

8 Conclusion

This paper combines a controlled VLM benchmark with an interactive expert-review artifact for open-set tattoo analysis. The benchmark shows that Qwen2.5-VL-7B achieves the strongest overall balance, whereas Gemma3 falls back to *unknown* more frequently and LLaMA substantially overpredicts under reduced-context conditions. The black/white crop comparison further demonstrates that visually irrelevant background changes can alter model outputs.

TattooAudit converts these discrepancies into a transparent review workflow. The implemented pipeline discovers benchmark cases, aligns reference and model artifacts, reconstructs missing case metrics when necessary, applies deterministic priority rules, joins the latest human-review status, presents visual and cross-model evidence, validates structured expert actions, preserves the decision history, and exports review records. The revised design positions the specialist as the final authority and treats VLM outputs as supporting evidence.

The current prototype begins after the offline pipeline has generated mask-derived crops, VLM outputs, and metrics. A future end-to-end system could start from original-image repositories, but it would need a preceding mask-acquisition and model-inference component that the present study neither

implements nor evaluates.

The system does not claim to automate forensic judgment, infer cultural or criminal meaning, establish demographic fairness, or demonstrate usability with domain experts. It also does not yet implement the governance floor required for sensitive operational categories. Its contribution is a functional and technically explicit bridge between benchmark evaluation and traceable expert review. This foundation supports future user studies, longitudinal evaluation, category-stratified analysis, governance-aware routing, and the carefully curated use of adjudicated review records for model and dataset improvement.

Declarations

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Authors' Contributions

Ricardo Alexandre Vargas Pereira: Conceptualization, Methodology, Software, Data curation, Investigation, Formal analysis, Visualization, Writing – original draft. Clayton Kossoski: Conceptualization, Supervision, Validation, Writing – review & editing. Rodrigo Tchalski da Silva: Validation, Writing – review & editing. Heitor Silvério Lopes: Supervision, Writing – review & editing.

Competing interests

The authors declare that they have no competing interests.

Availability of data and materials

The public dataset analyzed in this study is available in the TSSD2023 repository at <https://github.com/Brilhador/tssd2023>. The artifact developed for this study—including the evaluation scripts, prompt templates, parsing rules, exported tables, case-ranking outputs, and TattooAudit source code—is publicly available at <https://github.com/professorvargas/tattoo-audit-artifact>.

Further relevant information

This study reused publicly available benchmark data and did not recruit human participants. A future user study with expert reviewers will require a dedicated protocol and, where applicable, prior ethics review under institutional procedures. The authors used generative AI tools during manuscript preparation to support language revision, structural refinement, and drafting. The authors critically reviewed and validated all technical decisions, analyses, interpretations, figures, and final textual revisions and take full responsibility for the manuscript's content.

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